1.3 Satisfiability

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Definition: A compound proposition is <u>satisfiable</u> if you can assign truth values which make it (T)rue. Otherwise it is <u>unsatisfiable</u>.

Can I assign a truth value to p to make $p \vee \sim p$ true?

what about $p \wedge \sim p$?

Consider $\sim(r \wedge p) \vee (q \oplus p) \vee (\sim r \wedge q)$

To be (T)rue, we need at least one piece to be true.

$\sim(r \wedge p) \overset{?}{\equiv} \mathbb{T}$

$\Rightarrow r \wedge p \equiv \mathbb{F}$

$\Rightarrow \boxed{r = \mathbb{F} ; p = \mathbb{F}}$

$q \oplus p \overset{?}{\equiv} \mathbb{T}$

$\Rightarrow \boxed{\begin{array}{c} q = \mathbb{T}, p = \mathbb{F} \\ \text{or} \\ q = \mathbb{F}, p = \mathbb{F} \end{array}}$

$\sim r \wedge q \equiv \mathbb{T}$

$\Rightarrow \sim r = \mathbb{T}, q = \mathbb{T}$

$\Rightarrow \boxed{r = \mathbb{F}, q = \mathbb{F}}$

Then we can choose any of these as long as we make it complete.

The assignment $p = \mathbb{F}, q = \mathbb{T}, r = \mathbb{F}$ makes the compound prop. (T)rue hence it is <u>satisfiable</u>.

If the original problem used conjunction <u>instead</u>

"$\sim(r \wedge p) \wedge (q \oplus p) \wedge (\sim r \wedge q)$"

we would need to do more.

Requires <u>each</u> piece to be (T)rue now.

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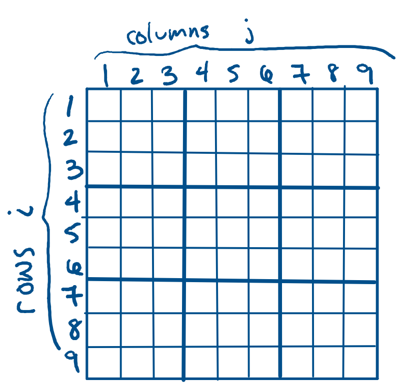
Split into 4 teams.
Are the following satisfiable?
If so, what truth values work?
If not, explain why.

1. $(p \vee \sim q) \wedge (q \vee \sim r) \wedge (r \vee \sim p)$
2. $(p \vee q \vee r) \wedge (\sim p \vee \sim q \vee \sim r)$
3. $(p \vee \sim q) \wedge (q \vee \sim r) \wedge (r \vee \sim p) \wedge (p \vee q \vee r)$

Satisfiability has applications in a wide variety of studies. In particular, it is used in programs which make and solve puzzles. The book gives a really nice description and story for the n-Queens Problem as well as Sudoku.

For sudoku we know we want 1,…,9 in each row column and 3x3 box without duplicates. We can encode this using compound propositions!



$n = 1, \ldots, 9$

If the (i,j) entry has value n, we will say $p(i,j,n) = \mathbb{T}$ else $p(i,j,n) = \mathbb{F}$.

Verifying row 1 contains the number 3
can be done by checking if

$$p(1,1,3) \vee p(1,2,3) \vee \cdots \vee p(1,9,3)$$

is satisfiable.

Remember, this can be shortened

$$\bigvee_{j=1}^{9} p(1,j,3)$$

To make sure each number appears

$$\left(\bigvee_{j=1}^{9} p(1,j,1)\right) \wedge \left(\bigvee_{j=1}^{9} p(1,j,2)\right) \wedge \cdots \wedge \left(\bigvee_{j=1}^{9} p(1,j,9)\right)$$

Since we want <u>all</u> of them to appear, we
need <u>conjunction</u>.

This simplifies to...

$$\bigwedge_{n=1}^{9} \bigvee_{j=1}^{9} p(1,j,n)$$

So then to make sure <u>every</u> row contains
every number, we form another <u>conjunction</u>
over all the rows.

$$\bigwedge_{i=1}^{9} \bigwedge_{j=1}^{9} \bigvee_{j=1}^{9} p(i,j,n)$$

This process is then done for
checking all the columns and
all the boxes.

Then with careful way of
encoding a sudoku puzzle,
we ask the program to solve
the <u>satisfiability</u> problem
which in turn solves the
sudoku.